

Unit6:Working with Real World Data

Real-World Social Data

Definition

- ▶ Social data is generated from online platforms such as:
 - ▶ Facebook, Instagram
 - ▶ X (Twitter)
 - ▶ YouTube, Reddit
 - ▶ TikTok
- ▶ It includes:
 - ▶ Text: posts, comments
 - ▶ Media metadata (images/videos)
 - ▶ Engagement: likes, shares, reactions
 - ▶ User interactions: followers, networks
 - ▶ Time-based data: timestamps

Key Characteristics of Social Data

- ▶ **Noisy:** slang, emojis, misspellings
- ▶ **Unstructured:** mostly text, images, videos
- ▶ **Large-scale:** millions of posts per day
- ▶ **Dynamic:** continuously changing in real time
- ▶ **Biased:** not fully representative of population

Data Pipeline in Social Data Applications (Part 1)

Step 1: Data Collection

- ▶ APIs (Twitter API, Reddit API)
- ▶ Web scraping
- ▶ Streaming real-time data

Step 2: Data Cleaning

- ▶ Remove duplicates
- ▶ Handle missing values
- ▶ Clean text (hashtags, URLs, emojis)
- ▶ Normalize (lowercase, stemming, lemmatization)

Data Pipeline in Social Data Applications (Part 2)

Step 3: Preprocessing

- ▶ Tokenization
- ▶ Stop-word removal
- ▶ TF-IDF / word embeddings

Step 4: EDA & Modeling

- ▶ EDA: hashtags, sentiment, trends, networks
- ▶ Modeling: sentiment analysis, topic modeling, spam detection, recommendations

Step 5: Visualization

- ▶ Word clouds, time-series plots
- ▶ Network graphs, heatmaps

Case Study: Social Media Sentiment Analysis

Overview

Social media platforms generate massive real-world data that can be analyzed to understand public opinion.

Objective:

- ▶ Analyze sentiment from social media data
- ▶ Classify opinions as Positive, Negative, or Neutral
- ▶ Track sentiment trends over time

Example Applications:

- ▶ Product launch feedback analysis
- ▶ Election sentiment monitoring
- ▶ Brand reputation analysis

Real-World Social Data Description

Data Sources:

- ▶ Facebook, Instagram
- ▶ X (Twitter)
- ▶ YouTube comments
- ▶ Reddit discussions
- ▶ TikTok engagement data

Data Includes:

- ▶ Text: posts, comments
- ▶ Media metadata (images/videos)
- ▶ Engagement: likes, shares, reactions
- ▶ User interactions: followers, networks
- ▶ Time-based activity (timestamps)

Key Characteristics of Social Data

- ▶ **Noisy:** slang, emojis, misspellings
- ▶ **Unstructured:** text, images, videos instead of tables
- ▶ **Large-scale:** millions of posts daily
- ▶ **Dynamic:** continuously changing in real time
- ▶ **Biased:** not fully representative of population

Data Collection Process

- ▶ Data extracted using APIs (Twitter API, Reddit API)
- ▶ Web scraping from social platforms
- ▶ Streaming real-time data feeds
- ▶ Keywords/hashtags used for filtering (e.g., #productname)
- ▶ Data stored in CSV, SQL, or NoSQL databases

Challenges:

- ▶ API rate limits
- ▶ Missing or incomplete data
- ▶ Spam and bot-generated content

Data Cleaning

Cleaning Steps:

- ▶ Remove duplicates and null values
- ▶ Clean text (remove URLs, hashtags, mentions)
- ▶ Handle emojis and special characters
- ▶ Normalize text (lowercasing, standardization)

Data Preprocessing

Text Processing Steps:

- ▶ Tokenization
- ▶ Stop-word removal
- ▶ Stemming / Lemmatization

Feature Extraction:

- ▶ TF-IDF vectors
- ▶ Word embeddings (Word2Vec, BERT)

Exploratory Data Analysis (EDA)

- ▶ Most frequent words and hashtags
- ▶ Sentiment distribution (Positive / Negative / Neutral)
- ▶ Trend analysis over time
- ▶ User engagement patterns

Visualization Methods:

- ▶ Word clouds
- ▶ Time-series plots
- ▶ Bar charts
- ▶ Heatmaps

Model Building

Machine Learning Models:

- ▶ Naive Bayes
- ▶ Logistic Regression
- ▶ Support Vector Machines (SVM)
- ▶ Deep Learning models (LSTM, BERT)

Task:

- ▶ Sentiment classification (Positive / Negative / Neutral)

Evaluation Metrics:

- ▶ Accuracy
- ▶ Precision, Recall, F1-score

Results and Insights

- ▶ Sentiment distribution varies by topic/event
- ▶ Major events cause spikes in sentiment changes
- ▶ Influential users significantly impact trends

Example Findings:

- ▶ 65% positive sentiment after product launch
- ▶ Negative spikes due to service issues or controversies

Challenges and Conclusion

Challenges:

- ▶ Noisy and unstructured text (slang, emojis)
- ▶ Fake accounts and bots
- ▶ Large-scale data processing
- ▶ Bias in user population
- ▶ Privacy and ethical concerns

Conclusion:

- ▶ Social media data provides valuable real-time insights
- ▶ NLP enables sentiment extraction from unstructured data
- ▶ Useful in marketing, politics, and decision-making

Key Idea: Transforming unstructured social data into actionable intelligence.

Example: Twitter Sentiment Analysis

Scenario: Analyzing public opinion on a new smartphone launch using Twitter data.

Sample Tweets:

- ▶ "Absolutely loving the new camera quality! NewPhone"
- ▶ "Battery drains too fast, very disappointing."
- ▶ "It's okay, but not worth the price."

After Sentiment Analysis:

- ▶ Positive: 1st tweet
- ▶ Negative: 2nd tweet
- ▶ Neutral: 3rd tweet

Insight: Majority sentiment helps companies understand product reception in real time.

Data Science Pipeline for Social Data

Overview

The social data pipeline transforms raw, unstructured social media data into meaningful insights for decision-making.

Main Goal:

- ▶ Extract useful information from social platforms
- ▶ Support decision-making using data-driven insights

Common Applications:

- ▶ Sentiment analysis
- ▶ Trend detection
- ▶ Recommendation systems
- ▶ Spam/bot detection

Data Collection and Storage

Data Collection Sources:

- ▶ APIs (Twitter/X, Reddit, YouTube)
- ▶ Web scraping
- ▶ Real-time streaming data

Data Types:

- ▶ Text (posts, comments, tweets)
- ▶ Metadata (time, location, user info)
- ▶ Engagement data (likes, shares, retweets)
- ▶ Network data (followers, interactions)

Data Storage:

- ▶ SQL databases (structured data)
- ▶ NoSQL databases (MongoDB)
- ▶ Data lakes (Hadoop / cloud storage)

Data Cleaning and Preprocessing

Data Cleaning:

- ▶ Remove duplicates and missing values
- ▶ Remove URLs, hashtags, mentions
- ▶ Handle emojis and special characters
- ▶ Normalize text formats

Text Preprocessing:

- ▶ Tokenization
- ▶ Stop-word removal
- ▶ Stemming / Lemmatization

Feature Extraction:

- ▶ TF-IDF representation
- ▶ Word embeddings (Word2Vec, GloVe, BERT)

EDA and Modeling

Exploratory Data Analysis (EDA):

- ▶ Word and hashtag frequency analysis
- ▶ Sentiment distribution
- ▶ Trend analysis over time
- ▶ User engagement patterns

Modeling Tasks:

- ▶ Sentiment analysis
- ▶ Topic modeling
- ▶ Spam/bot detection
- ▶ Recommendation systems
- ▶ Trend prediction

Algorithms:

- ▶ Naive Bayes, Logistic Regression
- ▶ SVM, Random Forest
- ▶ Deep learning (LSTM, BERT)

Evaluation and Insights

Model Evaluation Metrics:

- ▶ Accuracy
- ▶ Precision, Recall
- ▶ F1-score
- ▶ ROC-AUC

Insight Generation:

- ▶ Public sentiment trends
- ▶ Emerging topics and hashtags
- ▶ Influencer identification
- ▶ Behavior and engagement patterns

Visualization and Deployment

Visualization:

- ▶ Word clouds
- ▶ Line graphs (trend analysis)
- ▶ Bar and pie charts
- ▶ Heatmaps and network graphs

Deployment:

- ▶ Real-time dashboards
- ▶ Social media monitoring systems
- ▶ Recommendation engines
- ▶ Alert systems for trends or crises

Summary

Overall Pipeline:

Data Collection → Storage → Cleaning → Preprocessing → EDA
→ Modeling → Evaluation → Insights → Deployment

Key Idea:

Social data pipelines convert unstructured human-generated data into structured, actionable intelligence for decision-making.

Deployment using Streamlit

What is Deployment?

- ▶ Making ML models available for real users
- ▶ Turning code into interactive applications

Streamlit:

- ▶ Python framework for building web apps
- ▶ Used for ML model deployment and dashboards

Streamlit Deployment Workflow

- ▶ Train ML model using dataset
- ▶ Save model using pickle/joblib
- ▶ Build Streamlit interface
- ▶ Take user input via widgets
- ▶ Generate predictions in real time
- ▶ Display results on web UI

Key Features of Streamlit

- ▶ Simple Python-based framework
- ▶ No need for HTML/CSS/JS
- ▶ Real-time interactivity
- ▶ Fast prototyping of ML apps

Common Widgets:

- ▶ `st.text_input()`
- ▶ `st.button()`
- ▶ `st.slider()`
- ▶ `st.write()`

Example: Sentiment Analysis App

Workflow:

- ▶ User enters text
- ▶ Model processes input
- ▶ Prediction is generated

Output Classes:

- ▶ Positive
- ▶ Negative
- ▶ Neutral

Advantages and Limitations

Advantages:

- ▶ Easy and fast deployment
- ▶ No frontend skills required
- ▶ Great for ML demos

Limitations:

- ▶ Not suitable for large-scale production
- ▶ Limited customization
- ▶ Performance issues for heavy apps

Streamlit Sentiment Analysis App

Simple Real-World Deployment Example

This example demonstrates a Streamlit-based sentiment analysis application.

Streamlit App Code (Part 1)

```
import streamlit as st
import pickle

# Load trained model (example)
model = pickle.load(open("model.pkl", "rb"))
vectorizer = pickle.load(open("vectorizer.pkl", "rb"))

# Title
st.title(" Sentiment Analysis App")

st.write("Enter a text and check whether it is Positive, Ne
```

Streamlit App Code (Part 2)

```
# Input
user_input = st.text_input("Enter your text:")

# Prediction
if st.button("Predict Sentiment"):

    if user_input:
        # Transform input
        data = vectorizer.transform([user_input])

        # Predict
        prediction = model.predict(data)
```

Streamlit App Code (Part 3)

```
# Output
if prediction[0] == 1:
    st.success(" Positive Sentiment")
elif prediction[0] == 0:
    st.error(" Negative Sentiment")
else:
    st.info(" Neutral Sentiment")

else:
    st.warning("Please enter some text!")
```


How to Run the App

Run the Streamlit application using:

```
streamlit run app.py
```

Output:

- ▶ Opens a local web browser interface
- ▶ User enters text
- ▶ Model predicts sentiment in real time

Flask for Machine Learning Deployment

What is Flask?

Flask is a lightweight Python web framework used to build web applications and APIs.

In Data Science, Flask is used for:

- ▶ Deploying machine learning models
- ▶ Creating REST APIs
- ▶ Connecting frontend with ML backend

Why Flask for ML Deployment?

- ▶ Simple and easy to use
- ▶ Lightweight framework
- ▶ Flexible for API development
- ▶ Integrates with ML libraries (Scikit-learn, TensorFlow)
- ▶ Suitable for production-level backend services

Machine Learning Deployment Pipeline

- ▶ Train machine learning model
- ▶ Save model using pickle / joblib
- ▶ Load model inside Flask application
- ▶ Create API endpoints using routes
- ▶ Accept user input (text/data)
- ▶ Return prediction output

Core Components of Flask App

1. Flask App

- ▶ Created using `Flask(name)`

2. Routes

- ▶ Define URLs using `@app.route()`
- ▶ Example: `/predict`, `/home`

3. Request Handling

- ▶ GET requests (URL input)
- ▶ POST requests (form/JSON input)

4. Response

- ▶ HTML page or JSON output

Example: Sentiment Analysis API

Input:

- ▶ "This product is amazing!"

Processing:

- ▶ Text preprocessing
- ▶ Feature extraction (TF-IDF / embeddings)
- ▶ Model prediction

Output:

- ▶ Positive
- ▶ Negative
- ▶ Neutral

Flask vs Streamlit

Feature	Flask	Streamlit
Type	Web framework	ML app framework
UI	Manual (HTML/CSS)	Auto-generated UI
Complexity	Medium	Easy
Use Case	APIs, production apps	Prototypes, dashboards

Advantages and Limitations

Advantages:

- ▶ Lightweight and flexible
- ▶ Full control over backend
- ▶ Ideal for REST APIs
- ▶ Widely used in industry

Limitations:

- ▶ Requires HTML/CSS for UI
- ▶ More setup than Streamlit
- ▶ Not ideal for rapid prototyping

Summary

Key Idea:

Flask connects machine learning models to web applications using APIs.

Overall Insight:

- ▶ ML model → Flask API → User Interface

Ethical Considerations in Data Science

Definition:

Ethics in Data Science refers to responsible collection, analysis, and use of data while respecting privacy, fairness, transparency, and accountability.

Why it is important?

- ▶ Data affects individuals and society
- ▶ Incorrect use can lead to harm or discrimination

Data Privacy

Meaning: Protecting personal and sensitive user information.

Issues:

- ▶ Unauthorized data collection
- ▶ Data leaks and breaches
- ▶ Tracking without consent

Solutions:

- ▶ Data anonymization
- ▶ Encryption techniques
- ▶ Collect only required data

Bias and Fairness

Meaning: Bias occurs when data or models produce unfair outcomes.

Types of Bias:

- ▶ Sampling bias
- ▶ Algorithmic bias
- ▶ Historical bias

Impact:

- ▶ Discrimination in hiring, loans, and services

Transparency and Explainability

Problem: Many ML models are “black boxes”.

Solutions:

- ▶ Use interpretable models when possible
- ▶ Apply explainability tools (SHAP, LIME)
- ▶ Provide clear documentation

Data Security and Consent

Data Security:

- ▶ Protect data from unauthorized access
- ▶ Prevent cyberattacks and data theft

Consent and Ownership:

- ▶ Users must agree to data usage
- ▶ Transparent data policies are required

Accountability and Misuse of Data

Accountability:

- ▶ Responsibility for model decisions
- ▶ Human oversight in critical systems

Misuse of Data:

- ▶ Fake news and misinformation
- ▶ Deepfakes
- ▶ Surveillance misuse

Social Impact and Summary

Positive Impacts:

- ▶ Better healthcare and predictions
- ▶ Fraud detection systems

Negative Impacts:

- ▶ Job displacement
- ▶ Privacy risks
- ▶ Social manipulation

Key Idea:

- ▶ Ethical data science ensures fairness, transparency, and responsibility

Thank You

Thank You!